Weekly Homework 02: Data-Driven Programs

Course: Python Programming

Textbook: Murach's Python Programming, 2nd edition

Related class activity: Activity 2, Trivia Night

Suggested time: 2-3 hours

Total: 100 points

Submission clarification: October 11, 2026

Purpose

In class, you built a trivia game that stores questions as structured data and processes them with functions and loops. This homework gives you individual practice with lists, dictionaries, functions, and input validation using the textbook's exercises.

Read and complete: all three exercises are required

Use the exercise files that accompany your textbook, if provided. Follow the book's instructions for opening and running each program. The page numbers below are the printed page numbers in the 2nd edition.

A. Required: Dictionaries - Exercise 12-2, pages 353-354

Complete steps 3-7 of "Enhance the Movie List 2D program." Convert each movie record to a dictionary while keeping the collection of movies as a list. Update the list, add, and delete behavior so the program still works.

This is the main assignment because it practices the same data shape used by the trivia question bank: a list containing dictionaries with named fields.

B. Required: Lists and Functions - Exercise 6-1, pages 199-200

Exercise 6-1 is required. Complete steps 3-6 of "Use a list for the Test Scores program." Store the scores in a list, pass that list between functions, and calculate the total, count, and average.

Only step 7's additional statistics are optional enrichment. This does not make Exercise 6-1 optional.

C. Required: Exceptions - Exercise 8-1, page 251

Complete steps 3-4 of "Add exception handling to the Future Value program." Test invalid numeric entries, add exception handling, and test again. Pay attention to the difference between invalid text that cannot be converted to a number and a number that is outside the allowed range.

What to submit: Canvas code and journal, not Git

Submit the completed Python files for **all three exercises** through Canvas. Use clearly named individual .py files, or a folder/ZIP if Canvas accepts it.

Submit **one homework journal**, organized by exercise, as Canvas text or an uploaded document when that file type is accepted. If Canvas only accepts .py uploads, submit the journal in the Canvas text entry or submission comments. Do not publish homework to Git to work around upload restrictions.

Include the following in your journal:

- Your name, exercise numbers, filenames, and how to run each program.

- What you changed or tried in each exercise and your key takeaways.

- A short description of one problem you found and how you fixed it. If no problem occurred, explain a check you performed and what it showed.

- Testing evidence for the exercises: ordinary input and a relevant invalid-input or boundary case where applicable, including the input/action and the observed result. No screenshots or exact-output log format is required.

- An **AI-use note**: name the tool(s), explain what you requested, identify a suggestion you accepted, changed, or rejected, and explain how you verified the advice. If you did not use AI, state that.

No README is required. No Git link or Git publication is required for homework. Git publication is for the in-class labs. All homework explanations, run information, test notes, and AI-use notes belong in the journal; a separate document for each is not required.

Do not submit the textbook PDF, a classmate's code, passwords, API keys, or private information. Keep the original textbook exercise starter files intact by working on copies.

Using AI for homework

AI is allowed as a learning and review tool. You are responsible for the submitted work and must be able to explain it.

Good requests include:

- "Explain what this traceback is telling me. Give me one hint, not a complete rewrite."

- "Review my function call and explain why the value is not being returned as expected."

- "What tests should I try for this input validation? Do not write the solution."

Do not ask AI to complete the textbook exercises for you or submit code you cannot explain. Verify any generated advice by running your program and checking the result. Disclose meaningful AI assistance in your **journal**.

Rubric: 100 points

Criterion	Points	What earns credit
Exercise 12-2: list of dictionaries	30	Each record uses the required dictionary fields; list, add, and delete operations continue to work.
Exercise 6-1: list and function flow	25	Scores are collected in a list, passed to the processing function, and used to calculate correct total, count, and average.
Exercise 8-1: exception handling	20	Invalid number text is handled without an unexpected crash; valid and invalid entries are tested; range validation remains correct.
Testing and correctness	10	Programs run; journal notes cover normal and relevant edge cases; reported output is consistent with the code.
Journal	10	Meaningful account of exercise work, observations, takeaways, and debugging or a successful check. No unposted length or entry format.
AI-use disclosure and explanation	5	AI use is honestly disclosed, or non-use is stated, in the journal; the student can describe the submitted work.
Total	**100**	

Partial credit is available when a program is incomplete but the work demonstrates a correct attempt. A copied or unexplained solution may receive no credit for the affected criteria, consistent with course policy. Distinct journal, testing, and AI criteria may use evidence from the same journal; no duplicate documents are required.

One-time clarification for this already-assigned Homework 02: the instructor takes responsibility for inconsistent journal directions. No journal points will be deducted for this homework. Meaningful submitted reflection earns credit normally; otherwise the full 10 journal points are provided as an instructor allowance. This does not change the other rubric weights or retroactively add a journal format. For following homework, a journal will always be required, using the published standard homework expectations.

Before you submit

- Complete all three required exercises, including Exercise 6-1 steps 3-6.

- Run each program from a fresh start.

- Test both an ordinary case and at least one invalid or boundary case where relevant.

- Make sure your journal identifies the files and explains how to run them.

- Confirm you submitted your own work and included your AI-use or non-use note in the journal.
- Submit through Canvas, not Git. A README is not required.